Anti-Narcos Graph Intel

Graph-RAG Anti-Narcotics Intelligence Platform with Adaptive OSINT Crawling

Selected Track: Anti Narcotics

Team Name: The Conflicters

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Video Explanation: Click Here

Problem Statement

What is the problem? Why does it exist? Why is it important? Why solve it now?

1. What is the problem?

- Narcotics information is scattered across news, tips, raids, and media.
- People, places, drugs, and cases stay disconnected.
- Analysts cannot quickly see the full trafficking network.

2. Why does it exist?

- Data is siloed across agencies, portals, and formats.
- Tools search keywords, not relationships and patterns.
- No common loop links retrieval with live OSINT enrichment.

3. Why is it important?

- Missed links delay seizures, arrests, and disruption.
- Weak visibility lets networks reuse routes and aliases.
- Explainable, source-backed intel is critical for action.

4. Why does it need to be solved now?

- Trafficking networks move faster than manual workflows.
- Public OSINT volume is rising, but stays unstructured.
- AI + graph systems can convert signals into leads today.

Our Solution: Narco-Graph Intel

One adaptive loop: understand → retrieve → risk-check → enrich → answer

- Ingest news/manual text (multimodal conversion in roadmap).
- Convert to embeddings and retrieve via Graph-based Vector RAG.
- Start from high-similarity nodes and traverse with constraints.
- Mark suspicious graph patterns as HIGH RISK / HIGH PRIORITY.
- If high-confidence matches are low → free web search + crawl → store → re-query.

Operating Modes

Mode	When Used	Core Actions	Output
A · Retrieve	Enough high-similarity hits (≥ 0.80)	Top-K vectors + constrained Neo4j walk + risk checks	Top-N evidence + graph paths
B · Discover	High-confidence chunks below threshold	Search API → Scrapy extract → embed → graph upsert	Enriched KB + refreshed Top-N

Working / Approach

End-to-end system flowchart

1. Input
Text / news
(+ media later)

→

2. NLP
Entities &
keywords

→

3. Embed
Vectors &
chunks

→

4. Graph-RAG
Top-K +
constraints

→

5. Decide
Hits enough?
Risk flags?

Decision Branch

YES — Enough high-confidence matches

Constrained traversal → risk engine → compose Top-N output
(similarity ≥ 0.80, start from strongest seed nodes)

NO — Below threshold

Keyword search → crawl/extract → clean/embed → upsert Neo4j
then re-run Graph-RAG for enriched Top-N

Retrieval Policy Defaults

Top-K	Similarity Gate	Min High-Conf	Final Top-N	Graph Depth
100	≥ 0.80	10 chunks	10–20	2–3 hops

Technology Stack

Free / open-source preferred, mission-aligned tools

Layer	Tools	Role in Project	Cost
Backend / API	Python, FastAPI	Ingest, query, orchestration	Free
NLP	spaCy + lexicons	Entities: city, drug, raid, person	Free
Embeddings / RAG	sentence-transformers, LlamaIndex	Chunking, vectors, retrieval	Free
Graph DB	Neo4j Community (+ vector index)	Network map + Graph-RAG	Free
OSINT Crawl	SearXNG / CSE + Scrapy	Internet discovery fallback	Free
Multimodal*	Whisper, EasyOCR, FFmpeg	Audio/image/video → text	Free
UI / Viz	Streamlit/React + Cytoscape.js	Query console + graph view	Free

Innovation / USP

What makes this solution unique

Adaptive Knowledge Loop

If confidence is low, the system self-expands via ethical OSINT and re-answers.

Graph + Vector Fusion

Vectors find meaning; Neo4j reveals who/what/where is connected.

Constraint-Aware Search

Traversal expands only through useful constrained relationships.

Risk Prioritization

Suspicious patterns become high-priority alerts for human review.

Explainable Intel

Every answer keeps source URL, score, and graph evidence path.

Free Deployable Stack

Built to demo without paid API lock-in or enterprise licenses.

Target Users

Who benefits and how

User Segment	Primary Need	How Narco-Graph Helps
Law Enforcement Analysts	Link cases quickly	Graph paths + ranked evidence
Narcotics / Special Units	Prioritize leads	High-risk pattern alerts
OSINT Researchers	Structure public intel	Search→crawl→RAG pipeline
Situation / Command Rooms	Situational awareness	Top-N summaries with sources
Training Academies	Safe practice systems	Simulated public-data sandbox
Civic Tech Teams	Reusable architecture	Open-source Graph-RAG blueprint

Feasibility

Practical MVP boundaries for a winning build

Why It Is Feasible

- Core stack is mature and free/open-source.
- Neo4j supports relationships + vector retrieval.
- Scrapy + SearXNG cover public-web fallback.
- Text-first MVP can demo the full intelligence loop.
- Multimodal modules can be added without redesign.

MVP In-Scope vs Later

- In: text ingest, Graph-RAG, constraints, 1-2 risk rules.
- In: limited crawl fallback + basic UI.
- Later: full audio/video pipeline.
- Later: authorized special-source connectors.
- Always: public data + human review on alerts.

Feasibility Checklist

Tech Readiness	Legal/Public Scope	Demo Path Clarity	1–2 Day Full Vision
High	Controlled	Strong	No - use MVP cut

Scalability

Growth path after the first working system

Now
Single-node MVP
Seeded graph + text RAG + limited crawl

Next
Multi-source OSINT
More parsers, caching, better ranking

Scale
Hybrid data plane
Qdrant/Neo4j, workers, RBAC, alerts

Advance
Predictive intel
Route risk scoring + feedback learning

Dimension	Scale Lever	Outcome
Data	More sources + incremental embeddings	Broader recall, fresher knowledge
Product	Multi-analyst roles + case workspaces	Team usage in real workflows
Capability	Multilingual + predictive scoring	Deeper decision support

Timeline / Roadmap

Event build plan for a working demo

Phase	Focus	Deliverables	Exit Criteria
1	Foundation	Setup, Neo4j schema, seeded cases, text API	Data loads cleanly
2	Graph-RAG Core	Embeddings, Top-K, threshold filter, Top-N	Relevant retrieval works
3	Constraints + Risk	City/drug constraints, 1–2 risk rules, graph UI	Path + alert visible
4	OSINT Fallback	Search → Scrapy → upsert → re-query	Low-hit case auto-enriches
5	Demo Polish	UI flow, sample queries, pitch, PDF export	3-minute jury story ready

Impact, Future Scope & Limitations

Honest evaluation close

Expected Impact

- Faster case/entity linking
- Better recall via adaptive OSINT
- Clearer priority with risk flags
- Source-backed explainable outputs
- Catch and visualize the network of drug trafficking in a faster way.

Future Scope

- Full image/audio/video pipeline
- Strong EN/HI multilingual support
- Predictive route-risk scoring
- Authorized special-source connectors

Limitations & Challenges

- Not a whole-web indexer
- OCR/ASR/news parse noise
- Scraper breakage over time
- Alerts need human validation

One-line definition

Anti-Narcos Graph Intel converts multimodal/public signals into Graph-RAG intelligence, flags high-risk network patterns, and auto-expands knowledge through ethical OSINT when evidence is insufficient.